

SatsPath Architecture

The key-transparency extension is specified in Key Transparency Architecture. Resolution is not fully verified solely because a profile self-signature validates.

Overview

SatsPath is a routing and resolution layer that sits above existing Bitcoin payment protocols. It does not replace Lightning, Ark, or on-chain payments — it resolves human-readable identifiers to signed payment profiles and selects the best available rail.

System Diagram

Human-readable identifier
(e.g. rodrigo@satspath.dev)

|
v

Resolver /	
Transparent	< transactional SQLite event/profile/checkpoint state
Resolver	< pinned log_id + checkpoint/operator continuity
	< BIP-353 / HTTPS / Nostr / optional P2P

|
v

Signed Payment
Profile

- alias
- identity_pubkey
- methods[]
- signature

|
v

Merkle + history
+ ownership gate

v

Route Engine

		\
v	v	v

LN	BTC	Ark
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v	v	v

Invoice	PSBT	Ark Intent
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Crate Structure

```
satspath/  
  crates/  
    satspath-core/      # Types, crypto, codec, registry  
      profile.rs        # PaymentProfile, PaymentMethod, SignedPaymentProfile  
      crypto.rs         # secp256k1 keypair, sign, verify, fingerprint  
      codec.rs          # encode/decode satspath: URIs  
      registry.rs       # local file registry  
      errors.rs         # SatsPathError enum  
  
    satspath-router/    # Route selection engine  
      router.rs         # select_route() - priority logic  
      fees.rs           # mempool.space fee API client  
      lightning.rs      # Lightning availability + fee helpers  
      onchain.rs        # On-chain fee math + availability  
      ark.rs            # ArkClient trait + MockArkClient  
  
    satspath-cli/       # User-facing binary  
      commands/         # One module per CLI subcommand
```

Data Flow

Payment request lifecycle

1. Sender knows: rodrigo@satspath.dev + 21000 sats
2. Encode (optional):
satspath:v1:<base64url({"version":1,"alias":"rodrigo@satspath.dev","amount_sats":21000,...})>
3. Decode:
PaymentRequest { alias, amount_sats, memo, ... }
4. Resolve and verify:
profile -> signed event -> checkpoint-bound inclusion -> pinned consistency
-> operator continuity -> ownership-verified method allow-list
6. Route:
select_route(RouteRequest { alias, amount_sats, signed_profile })
-> RouteQuote { selected_method, reason, fee, confirmation }
7. Pay (simulated):
Execute against selected rail (Lightning invoice, on-chain tx, Ark intent)

Invite lifecycle (unregistered receiver)

1. Sender tries to pay julian@example.com
2. Registry returns AliasNotFound
3. create_invite("julian@example.com", amount_sats) -> Invite
4. Invite contains alias_hash, claim_url, warning
5. Sender shares claim_url with receiver out-of-band
6. Receiver generates their own keys locally and registers
7. Sender retries payment

Security Boundaries

- Private keys never leave the user's machine.
- The registry stores only public data (pubkeys, addresses, signatures).
- Profile, history, checkpoint binding, pin/operator continuity and selected method ownership verification happen before any quote/pay route.
- The `.satspath/` directory is git-ignored.

Pluggability

The registry is an abstraction — in production, `Registry::open()` can be swapped for: - BIP-353 DNS TXT record lookup - Nostr NIP-05 / NIP-57 resolution - Lightning Address `.well-known/lnurlp` discovery - A decentralized DHT

The router is similarly pluggable: new payment rails are added by implementing the routing priority logic and returning a `RouteQuote`.